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**B.TECH.- THIRD SEMESTER  
END-SEMESTER EXAMINATION, November-December  
2025**

Course Code-ICMTC301/EEMTC301/CMMTTC301  
Course Title-Engineering Mathematics

Time: 3:00 hours

Max Marks-50

*Note: Attempt all questions in given order only. Missing data/information (if any), maybe suitably assumed & mentioned in the answer.*

| Q. No. | Question                                                                                                                                                                                                                                                                                                                               | Marks | CO  |    |    |    |   |   |     |   |    |    |    |    |    |   |     |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|----|----|----|---|---|-----|---|----|----|----|----|----|---|-----|
| 1(a)   | Find the Fourier series expansion of the function $f(x) = x \sin x$ in the interval $(-\pi, \pi)$ and deduce that $\frac{\pi-2}{4} = \frac{1}{3} - \frac{1}{15} + \frac{1}{35} - \frac{1}{63} + \dots$                                                                                                                                 | 5     | CO1 |    |    |    |   |   |     |   |    |    |    |    |    |   |     |
| 1(b)   | Find the half range sine series of the function $f(x) = x^2$ in the interval $0 < x < l$ .                                                                                                                                                                                                                                             | 5     | CO1 |    |    |    |   |   |     |   |    |    |    |    |    |   |     |
| 1(c)   | Compute the Fourier coefficients $a_0, a_1, a_2$ and $b_1, b_2$ in the expansion of the function tabulated as follows:<br><table><tr><td><math>x</math></td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td><math>y</math></td><td>9</td><td>18</td><td>24</td><td>28</td><td>26</td><td>20</td></tr></table> | $x$   | 0   | 1  | 2  | 3  | 4 | 5 | $y$ | 9 | 18 | 24 | 28 | 26 | 20 | 5 | CO1 |
| $x$    | 0                                                                                                                                                                                                                                                                                                                                      | 1     | 2   | 3  | 4  | 5  |   |   |     |   |    |    |    |    |    |   |     |
| $y$    | 9                                                                                                                                                                                                                                                                                                                                      | 18    | 24  | 28 | 26 | 20 |   |   |     |   |    |    |    |    |    |   |     |
| 2(a)   | Verify Stoke's theorem for $\vec{F} = xy\hat{i} + y\hat{j} + z^2\hat{k}$ , on the surface of rectangular lamina bounded by $x = 0, y = 0, x = 1, y = 2, z = 0$ .                                                                                                                                                                       | 5     | CO2 |    |    |    |   |   |     |   |    |    |    |    |    |   |     |
| 2(b)   | Evaluate $\int_S \vec{F} \cdot \vec{N} \, ds$ where $\vec{F} = 2x^2y\hat{i} - y^2\hat{j} + 4xz^2\hat{k}$ and $S$ is the closed surface of the region in the first octant bounded by the cylinder $y^2 + z^2 = 9$ and the planes $x = 0, x = 2, y = 0$ and $z = 0$ .                                                                    | 5     | CO2 |    |    |    |   |   |     |   |    |    |    |    |    |   |     |
| 2(c)   | If $\vec{d}$ is a constant vector, then show that $\vec{d} \times (\vec{\nabla} \times \vec{r}) = \vec{\nabla}(\vec{d} \cdot \vec{r}) - (\vec{d} \cdot \vec{\nabla})\vec{r}$ .                                                                                                                                                         | 5     | CO2 |    |    |    |   |   |     |   |    |    |    |    |    |   |     |

|      |                                                                                                                                                                                                                                                                                                                                                                                       |   |     |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----|
| 3(a) | State convolution theorem and by using it find the inverse Laplace transform of $\frac{s}{s^2+4s+7}$ .                                                                                                                                                                                                                                                                                | 5 | CO3 |
| 3(b) | Solve the differential equation $y'' + 2y' + 5y = e^{-t} \sin t$ with $y(0) = 0$ and $y'(0) = 1$ .                                                                                                                                                                                                                                                                                    | 5 | CO3 |
| 3(c) | Find the Laplace transform of the function $f(t) = \frac{\cos \sqrt{t}}{\sqrt{t}}$ .                                                                                                                                                                                                                                                                                                  | 5 | CO3 |
| 4(a) | For $m, n \in \mathbb{N} \cup \{0\}$ , prove the following orthogonality property of the Legendre polynomials:<br>$\int_{-1}^1 P_m(x)P_n(x) \, dx = \begin{cases} 0, & \text{when } m \neq n \\ \frac{2}{2n+1}, & \text{when } m = n. \end{cases}$                                                                                                                                    | 5 | CO4 |
| 4(b) | Prove the following recurrence relation for the Legendre polynomials: $nP_n(x) = xP'_n(x) - P'_{n-1}(x)$ .                                                                                                                                                                                                                                                                            | 5 | CO4 |
| 4(c) | For the Legendre polynomials $P_n$ , show that $\int_{-1}^1 x^2 P_{n+1}(x)P_{n-1}(x) \, dx = \frac{2n(n+1)}{(2n-1)(2n+1)(2n+3)}$ .                                                                                                                                                                                                                                                    | 5 | CO4 |
| 5(a) | Find the solution of the Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ which satisfies the following boundary conditions:<br>(i) $u \rightarrow 0$ as $y \rightarrow \infty$ for all $x$ .<br>(ii) $u = 0$ at $x = 0$ for all $y$ .<br>(iii) $u = 0$ at $x = l$ for all $y$ .<br>(iv) $u = l(x-l)$ if $y = 0$ for all $x \in (0, l)$ . | 5 | CO5 |
| 5(b) | Solve the heat equation $\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$ , if<br>(i) $u(0, t) = 0$ ,<br>(ii) $u_x(l, t) = 0$ ,<br>(iii) $u(x, t)$ is bounded and<br>(iv) $u(x, 0) = \frac{y_0 x}{l}$ for $0 \leq x \leq l$ .                                                                                                                                     | 5 | CO5 |
| 5(c) | A string is attached and fastened to two points $l$ distance apart. Motion is started by replacing the string in the form $u = a \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$ . Find the displacement $u(x, t)$ from one end.                                                                                                                                      | 5 | CO5 |